OLAWALE AZEEZ

BACKEND & APPLIED AI ENGINEER | PYTHON · FASTAPI · RAG · CLOUD DELIVERY

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github.com/AZ1600 | az1600.github.io | Engineering Handbook | LinkedIn

PROFESSIONAL PROFILE

Python/FastAPI and cloud engineer with an IT support background and independently developed AI, API and platform systems. Designed and deployed a source-grounded RAG assistant on AWS with Markdown ingestion, embedding-based and keyword retrieval, cross-encoder reranking, structured evidence and automated evaluation. Additional engineering work covers PostgreSQL-backed application workflows, authentication and RBAC, Docker, Terraform, CI/CD, and operational telemetry. Interested in owning technically demanding backend and retrieval systems while retaining clear security and evaluation boundaries.

TECHNICAL EXPERTISE

Applied AI & retrieval: RAG pipelines; Markdown ingestion/chunking; Sentence Transformers; NumPy vector index; semantic + keyword hybrid retrieval; cross-encoder reranking; source metadata and no-evidence safeguards; LangGraph, MCP, Amazon Bedrock.

Backend, data & security: Python, FastAPI, Pydantic, REST APIs, PostgreSQL, SQL, boto3; authentication, JWT / OIDC, RBAC, managed identities, audit trails, tenant-aware application design.

Cloud delivery & operations: AWS ECS Fargate, ECR, ALB, IAM, CloudWatch; Azure Container Apps, Key Vault and Entra ID; Docker, Terraform, Bicep, GitHub Actions CI/CD, health checks and observability.

Engineering quality: Pytest, API and integration tests, retrieval evaluation cases, Git reviews, incident debugging, technical documentation and architecture trade-off analysis.

SELECTED INDEPENDENT ENGINEERING PROJECTS

RAG Operations Assistant — source-grounded knowledge retrieval GitHub

Python · FastAPI · NumPy index · Sentence Transformers · LangGraph · MCP · Amazon Bedrock · ECS Fargate

- Designed a Markdown document ingestion and chunking pipeline that generates normalized embeddings, a local NumPy vector index and JSON metadata recording source documents and chunk IDs.
- Combined semantic similarity and keyword scoring with cross-encoder reranking; returned source/chunk evidence and implemented a no-evidence path for unsupported questions.
- Built an evaluation suite covering top-1 source relevance, result-source purity and expected abstention, alongside mocked API tests and reusable LangGraph/MCP services.
- Deployed the FastAPI service to ARM64 AWS ECS Fargate behind an ALB using Terraform and GitHub Actions/OIDC; validated request flow, IAM scope, service stability and target health. Reduced documented compressed image size from ~3.5 GB to 586 MB using CPU-only PyTorch.

PropertyOps — secure backend workflows on Azure GitHub

FastAPI · PostgreSQL · Microsoft Entra ID · Azure Container Apps · Bicep · OpenTelemetry

- Developed maintenance-request triage, manager approval, persisted state transitions and audit history; enforced Entra application roles while keeping AI suggestions advisory.
- Deployed commit-SHA containers via OIDC-enabled GitHub Actions; integrated managed identities, scoped RBAC, Key Vault and Application Insights, with PostgreSQL persistence verified across revisions and restarts.

OpsPilot SaaS — multi-tenant operations workflow foundation GitHub

Next.js · TypeScript · PostgreSQL · Clerk · GitHub Actions

- Built tenant-scoped application workflows with server-side role checks, signal ingestion, human approval, execution states and audit records; separated recommendations from authorised actions.
- Used PostgreSQL-backed persistence, health/configuration checks and automated lint, type, unit and build validation; documented Vercel deployment and integration foundations.

FURTHER RELEVANT ENGINEERING WORK

CloudOps Command Center + PlatformPilot — evidence-led operational APIs GitHub

Next.js · TypeScript · FastAPI · Kubernetes · JSON Schema · Vitest

- Connected Kubernetes resource findings to owner-routed incident workflows with deterministic decision traces, runbooks, approval boundaries and persistent audit history.
- Defined and tested JSON integration contracts, authentication/permission edge cases, validation failures, TypeScript checks and production builds.

AI Resume Analyzer — browser-based document intelligence GitHub

JavaScript · PDF/DOCX/TXT parsing · deterministic scoring · n8n · Bedrock

- Built an interface that extracts résumé evidence locally in the browser, normalises role-related skills and gives explainable matches and gaps; added an optional n8n/Bedrock analysis flow.

Observability & Reliability Lab — production support practices GitHub

FastAPI · Prometheus · Grafana · OpenTelemetry · Loki · Tempo

- Instrumented API metrics, structured logs and traces; implemented latency/availability SLIs, SLOs and burn-rate alerts, then validated alert behaviour with deliberate failures and rule tests.

PROFESSIONAL EXPERIENCE

Junior IT Support Technician | HUMUCARE LTD • *Dec 2025 – Jul 2026*

- Investigated access, account, device and application issues; documented resolutions, escalated complex cases and communicated clear technical guidance to users.

Customer Service and Technical Support Operative | A&G Young Peth Grange • *2024*

- Captured and prioritised service and technical issues, coordinated appropriate escalation and kept stakeholders informed.

Customer Service Representative | Wemadek, Dubai • *2021 – 2023*

- Managed digital and telephone support cases, tracked follow-ups and worked with internal teams and suppliers to resolve issues.

PROFESSIONAL DOCUMENTATION & COLLABORATION

Cloud & Platform Engineering Handbook Read published site

Published project-linked cloud, AI/RAG and architecture guidance supported by implementation evidence; created documentation validation, internal-link checks and build automation.

CERTIFICATIONS & EDUCATION

AWS Certified Solutions Architect – Associate | AWS Certified Developer – Associate | AWS Certified Cloud Practitioner | AWS Certified AI Practitioner

MSc Biomedical Engineering, University of Dundee (UK) | BSc Biochemistry, University of Lagos (Nigeria)